

Product, Quotient, and Trigonometric Derivatives



The product rule

- 1 Differentiate $f(x) = (3x^2 - 1)(x^3 + 2x)$ using the product rule.
 - 2 Differentiate $f(x) = x^2 \sin x$.
 - 3 Differentiate $f(x) = e^x \cos x$.
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The quotient rule

- 4 Differentiate $f(x) = \frac{3x-1}{x+4}$.
 - 5 Differentiate $f(x) = \frac{x^2}{x^2+1}$.
 - 6 Differentiate $f(x) = \frac{\sin x}{x}$.
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Derivatives of the six trig functions

- 7 State the derivative of each trig function:

a $\frac{d}{dx} \sin x$

b $\frac{d}{dx} \cos x$

c $\frac{d}{dx} \tan x$

d $\frac{d}{dx} \cot x$

e $\frac{d}{dx} \sec x$

f $\frac{d}{dx} \csc x$

- 8 Differentiate $f(x) = x \tan x$.
 - 9 Differentiate $f(x) = \sec x \cdot \csc x$.
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Combining product, quotient, and trig rules

- 10 Differentiate $f(x) = \frac{e^x}{\sin x}$.
 - 11 Find the equation of the tangent line to $f(x) = x \cos x$ at $x = 0$.
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Derivatives using a table of values

- 12 Given $f(3) = 2$, $f'(3) = -1$, $g(3) = 4$, $g'(3) = 5$, find $h'(3)$ if $h(x) = f(x)g(x)$.

- 13 Using the same table, find $k'(3)$ if $k(x) = \frac{f(x)}{g(x)}$.
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Second derivatives with product and quotient rules

- 14 Find $f''(x)$ for $f(x) = x \sin x$.